R, RStudio, and the console · vectors · types

Tuesday, May 19, 2026

> Today
>
> **Before class**
>
> - **First Perusall Assignment Due**: *Data Computing* §2.1-2.8.
>
> **Plan for today**
>
> - Tour of RStudio
> - Using R as a calculator
> - Data types and vectors
> - Creating objects
>
> **Helpful links**
>
> - Syllabus · AI policy · Schedule · *Data Computing* §2

What is Computing?

Computing is about turning human ideas into reproducible instructions that a computer can carry out

In this course, we will learn how to:

- organize data
- automate calculations
- visualize information
- build reproducible analyses
- communicate results

R is the tool we will use to do that.

Another Introduction to R

- R is a programming language designed for data analysis, statistics, and visualization.
- We use R to tell the computer exactly what computations we want to perform.
- R is widely used in:
  - statistics
  - data science
  - research
  - industry analytics
- R is, free, open-source, and highly extensible through thousands of packages that contain additional tools and functions.

RStudio Desktop (RStudio)

- First version released in 2011.
- *Integrated development environment (IDE)* provides additional tools and window management to R to help coders, statisticians, and data scientists use R more effectively.
- While R is required to run RStudio, you do not need to have RStudio to use R.

R vs. RStudio

- **R** is the programming language.
- **RStudio** is the environment/interface we use to work with R more comfortably.

RStudio's Four Pane Layout

Source Pane

Console Pane

Environment Pane

Output Pane - Files Tab

Output Pane - Plots Tab

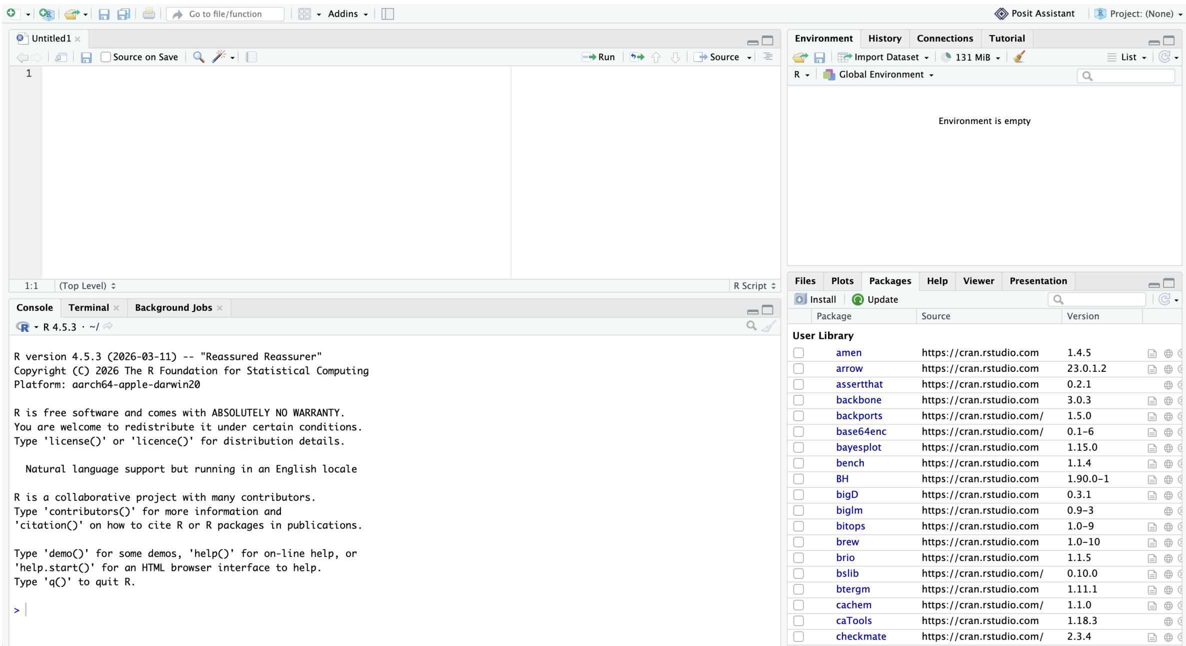


Figure 1: RStudio's four pane layout.

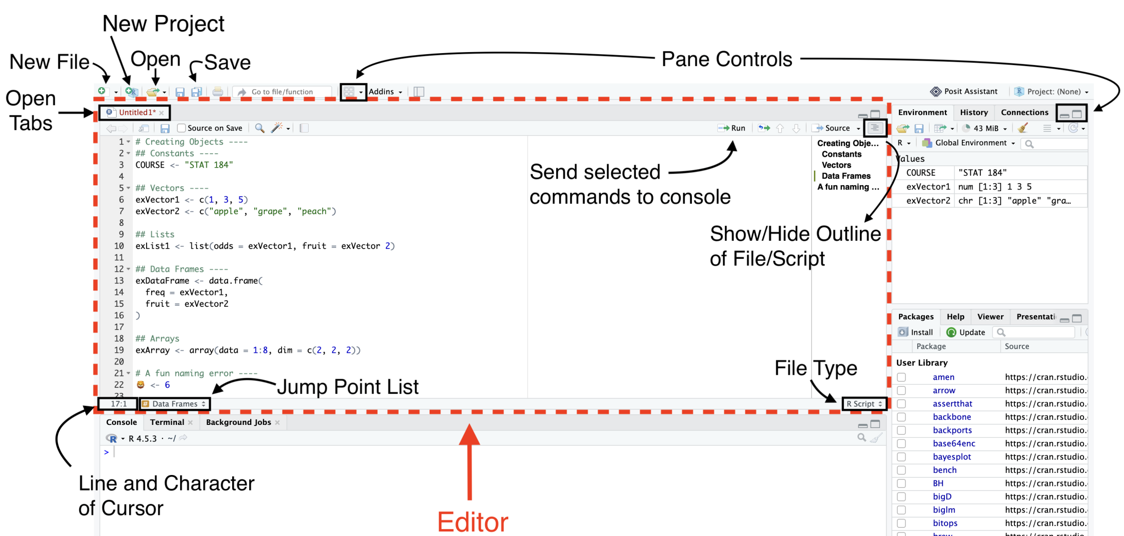


Figure 2: RStudio's source pane, annotated.

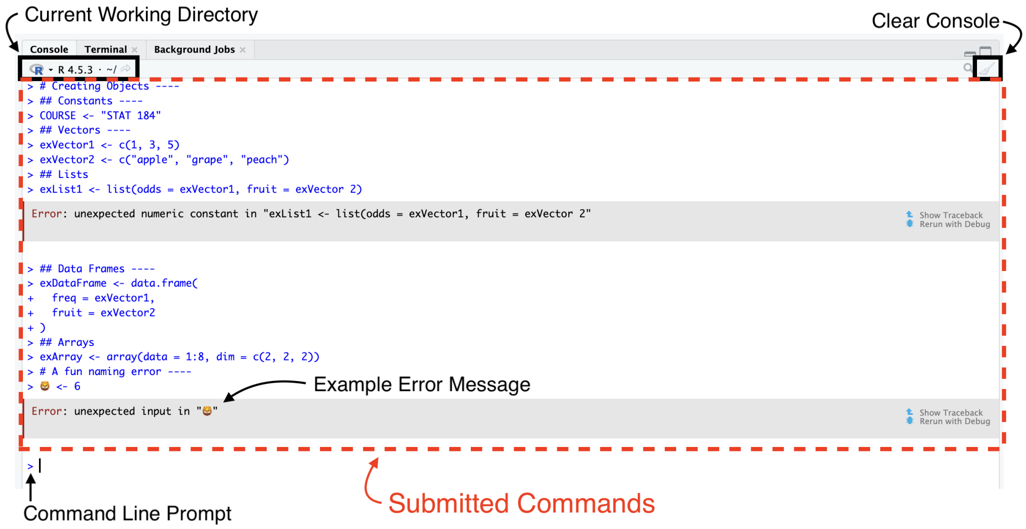


Figure 3: RStudio's console, annotated.

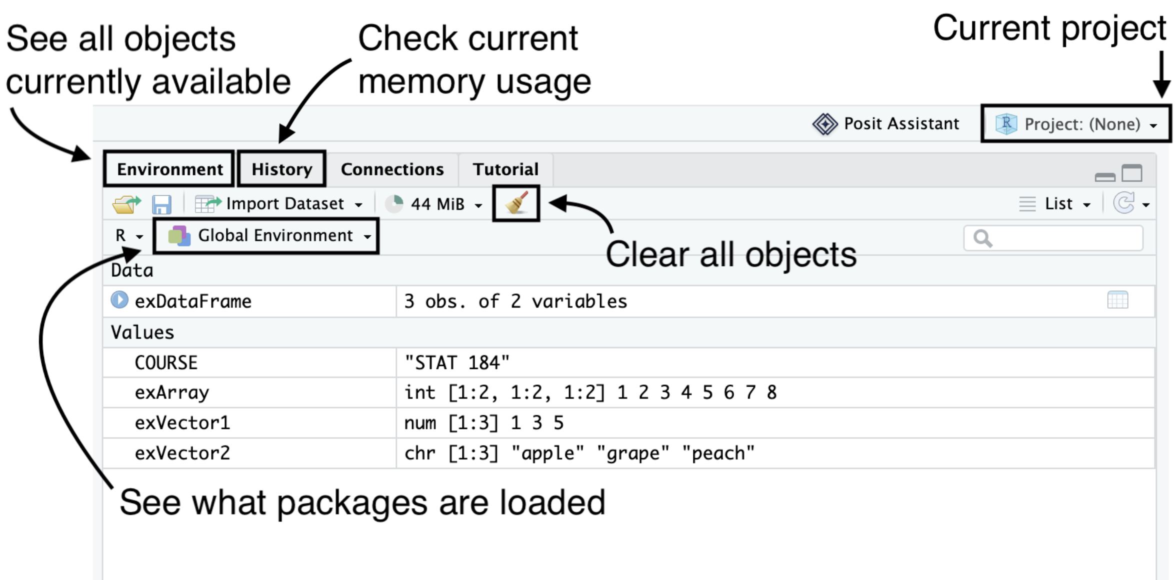


Figure 4: RStudio's environment pane, annotated.

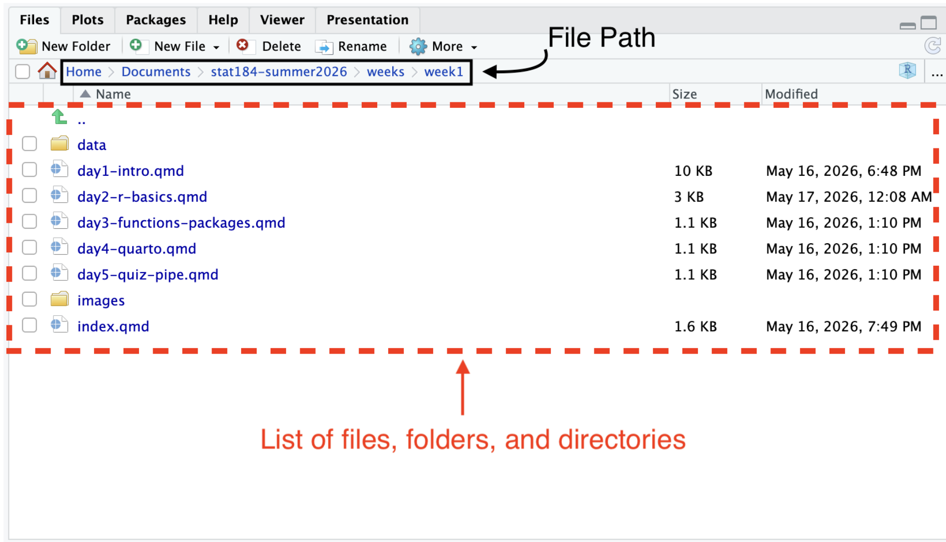


Figure 5: RStudio's output pane (files tab), annotated.

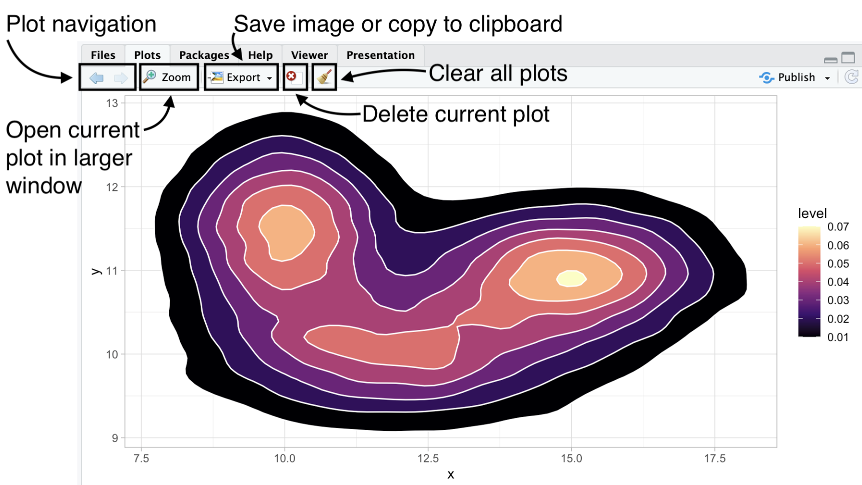


Figure 6: RStudio's output pane (plots tab), annotated, displaying a 2D density contour plot created with `ggplot2`. Darker regions indicate areas with higher concentrations of observations, while lighter regions represent lower densities. The plot combines filled density polygons with contour boundaries to visualize the distribution of simulated data.

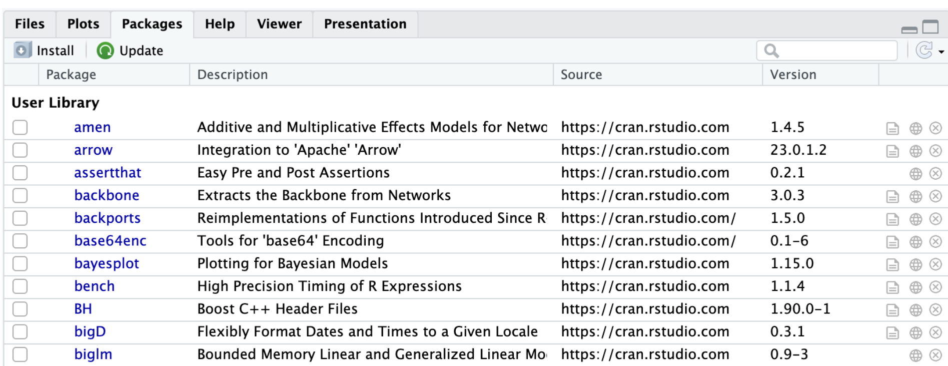
```r
# Set Seed
set.seed(184)

# Library
library(tidyverse)

# Data
a <- data.frame( x=rnorm(20000, 12, 2), y=rnorm(20000, 10, 0.5) )
b <- data.frame( x=rnorm(20000, 15, 1.5), y=rnorm(20000, 11, 0.5) )
c <- data.frame( x=rnorm(20000, 10, 1.1), y=rnorm(20000, 11.5, 0.7) )
data <- rbind(a,b,c)

# Area + contour
ggplot(data, aes(x=x, y=y) ) +
  stat_density_2d(aes(fill = ..level..), geom = "polygon", colour="white") +
  theme_light() +
  scale_fill_viridis_c(option = "magma")
```

Output Pane - Packages Tab



Package	Description	Source	Version
User Library			
amen	Additive and Multiplicative Effects Models for Netwo	https://cran.rstudio.com	1.4.5
arrow	Integration to 'Apache' 'Arrow'	https://cran.rstudio.com	23.0.1.2
assertthat	Easy Pre and Post Assertions	https://cran.rstudio.com	0.2.1
backbone	Extracts the Backbone from Networks	https://cran.rstudio.com	3.0.3
backports	Reimplementations of Functions Introduced Since R	https://cran.rstudio.com/	1.5.0
base64enc	Tools for 'base64' Encoding	https://cran.rstudio.com/	0.1-6
bayesplot	Plotting for Bayesian Models	https://cran.rstudio.com	1.15.0
bench	High Precision Timing of R Expressions	https://cran.rstudio.com	1.1.4
BH	Boost C++ Header Files	https://cran.rstudio.com	1.90.0-1
bigD	Flexibly Format Dates and Times to a Given Locale	https://cran.rstudio.com	0.3.1
biglm	Bounded Memory Linear and Generalized Linear Mo	https://cran.rstudio.com	0.9-3

Figure 7: RStudio's output pane (packages tab).

- The "Packages" tab will list all packages that you have currently installed on your machine. You can load a package by clicking the box to the left of the name.
- You can also use the "Install" and "Update" buttons to help you get new packages and keep them current.

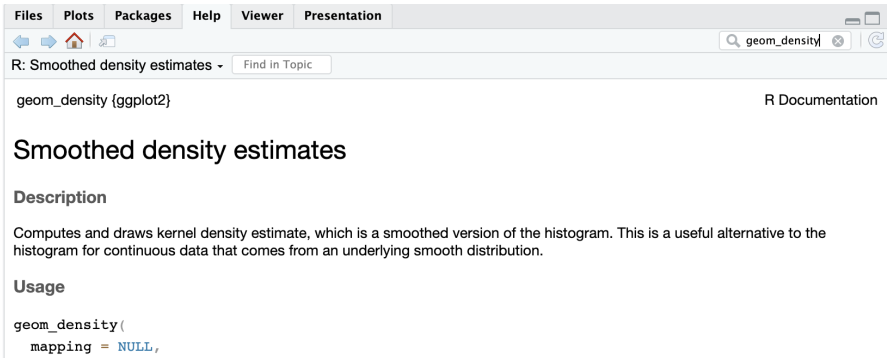


Figure 8: RStudio's output pane (help tab).

Output Pane - Help Tab

- The "Help" tab displays help documentation that has been included with the packages for functions, data sets, and other objects.
- **Help Shortcuts:** ?barplot, help(barplot), or ??barplot (broad sweep). You can also place your cursor in the object name and press fn + F1 (or just F1 if your function keys are active).

Data Types

- Data type refers to the nature of how we choose to record some value.
- **Common Data Types in R:**
  - Numeric (Double)
  - Integer
  - Complex
  - Character
  - Logical

Data Structures

- Data structures are the most important type of base objects in R as they will form the basis for all manipulations we do.
- R's data structures may be organized along two aspects:
  1. How many dimensions we need for values.
  2. If all values have to be the same data type.

Dimensions	All Values Same Data Type	A Mix of Data Types

1 Dimension	**Atomic Vectors**
	
	`[1] 1 3 5`
	`[1] "apple" "grape"`
	`"peach"`
		List
		
		`[[1]]`
		`[1] 1 3 5`
		
		`[[2]]`
		`[1] "apple" "grape"`
		`"peach"`

2 Dimensions	**Matrix**
	
	`     [,1] [,2]`
	`[1,]    1    5`
	`[2,]    3    7`
		Data Frame
		
		`  id  fruit`
		`1  1  apple`
		`2  3  grape`
		`3  5  peach`

n Dimensions	**Array**
	
	`, , 1`
	
	`     [,1] [,2]`
	`[1,]    1    5`
	`[2,]    3    7`
	
	`, , 2`
	
	`     [,1] [,2]`
	`[1,]    2    6`
	`[2,]    4    8`
		N/A

R as a Calculator

You can use R as a calculator:

```r
184.101 * 9.35
```

```
[1] 1721.344
```

```r
(8 + 4) / 9
```

```
[1] 1.333333
```

```r
sqrt(8)
```

```
[1] 2.828427
```

```r
cos(2*pi)
```

```
[1] 1
```

Vector Operations

```r
c(1, 8, 4) + c(9, 3, 5)
```

```
[1] 10 11  9
```

```r
c(1, 8, 4) * c(9, 3, 5)
```

```
[1]  9 24 20
```

- The `c` functions **concatenates** multiple individual values into one single vector.

Built-in Functions and Constants

```r
exp(1)
```

```
[1] 2.718282
```

```r
sqrt(pi)
```

```
[1] 1.772454
```

```r
dnorm(0)
```

```
[1] 0.3989423
```

- `sqrt` computes the square root.
- `exp` computes the function e^x.
- `pi` is the constant π.
- `dnorm` computes the standard normal density function.

Creating objects

You can create new objects with the **assignment operator**, `<-`

```
x <- 3 * 5
```

The above is an example of an **assignment statement**. In general, assignment statements have the form:

```
object_name <- value
```

In English, this is "object_name **gets** value."

- Even though typing `<-` for the many assignments you will make can be tedious, **don't use `=`**; it will work, but will cause confusion later.
- You can use the RStudio keyboard shortcut `Alt` + `-`, (or `Option` + `-` on a Mac).
- Object names must start with a letter, and can only contain letters, numbers, `_`, and `.`

Setup checklist — due by Wednesday

Install R and RStudio

- R: https://cloud.r-project.org
- RStudio: https://posit.co/download/rstudio-desktop

> ⚠ Mac users
>
> Install **XQuartz** too — some packages we'll use later in the course need it.

Configure RStudio

- `Tools > Global Options... > General`
- Restore .RData into workspace at startup: **Unchecked**
- Save workspace to .RData on exit: **Never**

Assignments due Wednesday 5/20

- HW 0: Setup confirmation

Wrap-up & looking ahead

Tomorrow: Functions, packages, and the tidyverse.

Due Wednesday before class:

- Read *Data Computing* §3.1–3.5 on Perusall.

Reference links

- Syllabus · AI policy · Schedule · *Data Computing*

> Reach out
>
> Stuck on setup? Come to office hours or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu) before Wednesday — don't wait until class.